11–13 OCTOBER 2026 · STAGES W0–W10 · BRANCH V2.1

# DegradX Experiment Log — v2.1

The third experiment programme, run under the brief “DegradX v2.1” on top of the frozen v2 results. It added a third evaluated profile, FORM, built from the TRI/Stanford formation dataset of Cui *et al.* (2024). FORM is the first profile whose generator inserts patterns. The one MATR and HUST computation of this programme measured the share of the correlated noise in transfer. The programme also delivered a review build of the paper in which every change since the authors' text is highlighted. ISU-ILCC stays closed under option A.

`3 profiles: MATR, HUST, FORM` `FORM: 182 cells, all reach EOL` `4 680 TimeSHAP explanations` `decisions D32–D34` `amendments round 8` `tags v2-results, declarations-frozen-r4`

RQ1 · FIDELITY (FORM)

**1.54** [1.28, 1.74]

FORM's transfer ratio. It lies outside HUST's interval [0.94, 1.34], as MATR's does, but the intervals overlap. Unlike MATR's, the gap is located: in the non-capacity channels, mostly in their early-life levels. FORM's discriminator cell is the first readable one (0.10–0.17).

RQ2 · USABILITY (FORM)

**3** of 3 · **0.0000**

Every weighted channel is used materially by every member (NRMSE 0.179). The pattern term is separable (|r| 0.045) but not shown to be learned: removing it leaves the error at 0.0000 [−0.0001, 0.0002], a test that cannot detect a term this small. So the trained-model sparse-set scores are void (D21 logic, no inflation).

RQ3 · SCORES

**1.00 → 0.37**

Paired retrieval of the undegraded map, then under the smallest added noise, measured for the first time. FORM reproduces the registering pattern of MATR and HUST exactly: rank agreement misses smoothing and partial shifts, and the temporal profile error registers them.

TRANSFER SHARES (MATR / HUST)

**2.19×** / 1.30×

The part of the v1 → v2 transfer gain carried by the per-unit offsets. Correlated noise carries 1.02× / 0.98×, i.e. none: the gains of v2 are the offsets' in every measure where the two were separated.

## Flags for the authors

These need a decision or a revision by the authors. The Notes box on the first page of `paper/review/DegradX_review.pdf` lists the same items, with pointers into the paper. None was settled by the agent.

1. **Licence.** The FORM profile and its generated units inherit CC BY-NC 4.0, a non-commercial term. The README says so. The cell-to-protocol sheet (the authors' GitHub file, no licence) is fetched and pinned, never redistributed.

2. **Nominal capacity of FORM**: 0.24 Ah, from Cui *et al.* p.3082 (“nominally 240 mAh … 1C is defined as 0.24 A”). BEEP's stored `nominal_capacity` agrees.
3. **HUST stays the anchor** (D29, the brief's ruling), on transfer, covariance and the property table, not the discriminator. FORM's transfer ratio, 1.54, lies outside HUST's interval, as MATR's does. The agent reports this and does not re-decide the anchor.
4. **Post-data rules and declaration timing.** v2.1 adds two post-data rules, both for FORM:
    - **D33**, the charge current of the protocol check read on constant-current samples. Read as declared, it would have stopped FORM on a false second protocol made by post-EOL collapsed cycles.
    - **A post-hoc sensitivity** of the mechanism diagnostic, which gives the same reading.

    r4 was written after every earlier result, and after FORM's documentation (with outcome-adjacent values) and the measured values of one of its cells had been read (all listed in `v2_1.exposure`), but before any download. All of this is in the Limitations, coloured.
5. **E3 = 66** corrects an off-by-one (67). Tongji is not re-opened.
6. **The D28 E2 reading for ISU-ILCC** is still a choice. ISU-ILCC is closed and reported, not evaluated.
7. **FORM's sparse set.** It exists, but no trained model is shown to learn it, so its retrieval is read on the reference model and on degraded maps only. Decide whether the paper presents it as a construction check only.
8. **The AI-use disclosure.** The Acknowledgments are untouched. A proposed disclosure is in the Notes box: the experiments were run by an AI agent, which also proposed the decision records and drafted the methodology amendments and the results text.
9. **The Abstract and Conclusion are agent drafts**, marked “DRAFT (agent) — for the authors to rewrite”. The repository and DOI footnote in §1 is still a placeholder.
10. **Table 1 layout.** r4 lists four TSTR rows for MATR/HUST. The paper prints “v2 (three seeds)” as the main row and “v2 without X3 (seed 0)” and “v1” as rows. “v2 (seed 0)” goes into the note.
11. **21 of 300 generated FORM units** end above the EOL threshold by more than the declared record-end tolerance (2.4 mAh), because their capacity noise varies slowly. They are left out of the property rows that depend on EOL (D30 applied as declared).

## v2 → v2.1: what changed

MATR and HUST values are the v2 values (tag `v2-results`, not re-run), except the TSTR rows on their X3-off sets. FORM values are from the W5–W8 artifacts. Recency weighting throughout.

| ITEM | V2 | V2.1 |
|---|---|---|
| Evaluated profiles | MATR, HUST | MATR, HUST, **FORM** (182 cells; 127 fitting / 55 held out) |
| Third dataset | ISU-ILCC, parked at V4 (offset share 0.77–0.82) | ISU-ILCC closed under option A; FORM evaluated (offset share 0.12 measured, 0.11–0.14 generated) |
| Inserted patterns | none enabled | FORM: post-diagnostic recovery enabled (2.28× noise rate); not shown to be learned by trained models |

| ITEM | V2 | V2.1 |
|---|---|---|
| Transfer ratio [95 % BCa] | MATR 1.49 [1.04, 2.29]; HUST 1.17 [0.94, 1.34] | unchanged; FORM 1.54 [1.28, 1.74] |
| Transfer without X3 (seed 0) | `not measured` | MATR 1.60 [1.06, 2.50]; HUST 1.12 [0.92, 1.29] |
| Shares in transfer: offsets / correlated noise | `not separable` | MATR 2.19× / 1.02×; HUST 1.30× / 0.98× |
| Discriminator cells | void (D02b) on both | MATR, HUST void; **FORM readable, 0.10–0.17** |
| Covariance gap | closed on both | closed on all three (FORM 0.35 within [0.20, 0.43]) |
| Primary NRMSE on y; channels used | 0.091 / 0.095; all, every member | FORM 0.179; all, every member, materially |
| Retrieval of the sparse set | `not computed` | FORM: paired retrieval on degraded maps (W7) and on the reference model (W8); trained models void |
| TimeSHAP explanations | 5 760 (MATR, HUST) | 4 680 (FORM, with pattern-free counterparts) |
| Methodology diagrams | the diagrams of the paper as given (Figure 4 redrawn at v1 S4), showing the generator without offsets or correlated noise | redrawn from scripts; every label from declarations or the FORM profile |
| Highlighting in the review build | methodology rounds 1–7 in `\rev` | every change since 6221c50 in §1–§3, two categories, checked mechanically; §4–§5 new with banners; Notes box |
| Abstract, Conclusion | template placeholders | agent drafts, marked for the authors |
| Declarations | r3 (`v2` block) | r4 (`v2_1` block), frozen at `declarations-frozen-r4` before download |

## Stage order

The stages ran and were committed in the order W0, W1, W2, W3, W4, **W7, W9, W6, W5, W8**, W10. W7, the degraded-map test, and W9, the diagrams, need no trained model, so they ran on the CPU while W6, the ensembles, and W5, the TSTR models, shared the GPU. W8 needs W6's models. Each stage reads only the artifacts of the stages it depends on, so the order changes no result. The sections below follow the stage numbers.

## W0 Documentation and declarations r4, before any download

`11 Oct · no data · HEAD requests only`

`commit 23ac263 · tag declarations-frozen-r4`

WHAT RAN

- **Branch and tag.** Tag `v2-results` on the v2 HEAD; branch `v2.1`; output roots `artifacts/v2.1`, `results/v2.1` and `data/v2.1`.

- **Documentation research.** Four readers, a synthesis of 72 claims in six groups, and 12 verifiers (source fidelity and an adversary per group). It covered Cui *et al.* and SI, Rhyu *et al.*, the data page, the authors' code and parameter sheet, BEEP, and BatteryLife (non-binding).
- **r4.** The block `declared_by_design.v2_1`: paths, authors' decisions, scope, eligibility (E3 = 66, E2 strict), archive rules, the pinned formation mapping, the cell scope, the aging protocol with its tolerance and second-protocol rule, positions and predicates, the three channels with the CC-CV charge-time definition and its validation, rest events, EOL reporting, converter checks, the mechanism diagnostic with its corrected start index (T7), the weight guard, the offset pre-screen, readings, 12 third-profile-only stop cases, `tstr_without_x3`, and exposure.
- **D32** settles what the documentation leaves open, before data. **T7** records the off-by-one in v2's ISU-ILCC mechanism diagnostic (closed, not recomputed).
- **Review before commit.** Three lenses (brief compliance, measurement science, pre-registration hygiene) and a judge. 57 objections (1 blocker, 27 major, 29 minor), all accepted: 12 as proposed, 45 modified.

---

Everything that governs FORM was committed and tagged before any byte of the archives was downloaded. What had been read by then, including outcome-adjacent values printed in the papers and the measured values of one cell, is listed in `v2_1.exposure`.

## W1 Acquisition and ingestion, with converter checks

12–13 Oct · 15.6 GB archive · 943 s conversion

commits `01889fd` hashes and member lists (before any member was opened) · `f818fda` ingestion, D33

### WHAT RAN

- **Fetch.** Size, ETag and Last-Modified match the declared HEAD values. SHA-256 and member lists were committed before extraction. The parameter sheet was fetched at its pinned commit and hash.
- **Extraction.** 182 aging files from the current archive, with the tarfile `data` filter. The original archive contributes nothing, since its 39 seqs are all in the current one.
- **Conversion.** `src/degradx/data/tri_formation.py` in two passes: per-cycle statistics, then the cross-cell rules.

### WHAT CAME OUT

- **182 of 182 cells in scope**, 189 485 aging positions (736–1 625 per cell). The modal protocol is 0.24 A / 0.18 A / 4.4 V / 3.0 V, as documented. 1 incomplete cycle in about 189 000.
- **BEEP.** Every file is from the post-2023 writer, with `full_fast_charge` = 1. This explains the brief's observation that `charge_duration` is NaN once a cycle charges less than 0.24 Ah.
- **Charge-time validation:** 178 of 178 evaluable files pass (4 not evaluable). BEEP's duration exceeds the reproduced CC-CV time by a near-constant 5.40–5.49 min, the zero-current segment at the cycle start (the documented reason).
- **Rest events:** 2 140 diagnostic, 88 paused and 112 gap events. **75 DST artefacts were set aside**: fall-back hours that BEEP maps as PDT.
- **All converter checks pass.** Summaries are equal to machine precision, the index structure is sound, and the diagnostics are in their documented layout in 543 of 546 (cell, type) pairs.

### D33 The protocol check reads the charge current on constant-current samples POST-DATA

FOUND Read as declared (the 90th percentile of all charge samples), 40 cells carry runs of more than 11 cycles with a departing charge current, agreeing in 114 pairs. That is a second aging protocol, which is a stop case. Every such cycle is long after EOL, with capacity at 0.006–0.046 Ah: the charge is almost all CV, so the percentile no longer measures the CC level. Discharge current and voltages are unchanged.

CHOSEN The median current over CC samples (below the cycle's own $V_{cv}$ − 10 mV, current above C/50, at least 10 samples). A cycle without a CC phase is not tested. Result: no cell out of tolerance, no second protocol.

KEPT The literal outcome is kept in `ingest_summary_r4_literal.json`, `cells_r4_literal.csv` and `checks_r4_literal.md`. Listed in the paper's Limitations and Notes box.

## W2 Audit, eligibility, mechanism diagnostic, offset pre-screen

13 Oct · CPU · split 127 / 55

commit e6d1d06 · D34

WHAT CAME OUT

| CRITERION | MEASURED | VERDICT |
|---|---|---|
| E3, units reaching EOL (≥ 66) | 182 of 182 (127 fitting, 55 held out); T median 689, min 463 | pass |
| E5, weight guard (\|Δφ\| / noise SD ≥ 1) | charge time 3.33; mean discharge voltage 5.86 | pass |
| E6, licence | CC BY-NC 4.0 | pass, NC flagged |
| Offset pre-screen, $Var(y_{offset})/Var(y)$ (stop > 0.5) | 0.120 (recency), 0.125 (uniform) | pass |
| Formation-protocol share of the offset term | $R^2$ 0.689 vs null 0.476, p = 0.0003; calibrated 0.41 | reported |
| D05, positive / negative runs vs noise | 2.28 (312 runs) / 0.96 | positive enabled |
| Mechanism, runs within 0–10 after a rest event | 4.57 [3.99, 5.18]; placebo 0.98 [0.63, 1.34] | localised after rests |
| Separability, 41 rests without a rate change | 2.04 [0, 4.51] | branch (iii): attributed to the diagnostic block |

**Post-hoc sensitivity (not declared).** The declared gap events mostly sit one position after a diagnostic block (112 events, median distance 1). The first aging cycle after a block charges from the RPT's deep discharge and lasts longer. Restricted to rests with no diagnostic in the 11 preceding positions (33 events), the ratio is 2.54 [0, 5.78]. The branch is the same, and the declared reading stands.

> **D34** **FORM is eligible, and the positive pattern type is enabled** PRE-DECLARED READINGS
>
> READING E1–E6 hold, the pre-screen reads 0.12, and D05 enables the post-diagnostic recovery. By the separability rule, the recovery is attributed to the diagnostic block (a rest and a lower-rate discharge), not to rest alone.

## W3 Profile fit under X2 and X3

13 Oct · CPU, 40 s · v2 V3 code
commit 47a6595

- **W2 reproduced exactly:** the same split, the two-term exponential family, and D05 positive enabled at 2.285.
- **Mappings:** charge time falls 2.91 min and mean discharge voltage 41.7 mV from z = 0 to 1. $x_0$ = 0.2459 Ah, 72.03 min, 3.738 V. The backfit converged in 4–5 iterations.
- **Within-unit residual correlation:** charge time against mean discharge voltage is **−0.89**, far beyond any MATR or HUST pair. Capacity against the others is about ±0.12. No positive-definite repair was needed.
- All checks pass. There are no redundant channels: temperature and internal resistance are absent.

## W4 Generated units: 3 × 300, plus X3 off

13 Oct · CPU, 95 s · first v2 profile with patterns
commit 3e7f635

- **Every construction check passes** in every seed and weighting, including “disabling patterns leaves z, T, R, the mean and the noise unchanged”.
- **Offset share** 0.115–0.137, far below 0.5.
- **Patterns are live but negligible in y:** 741–803 per seed in 213–216 units, amplitude median 2.7 mAh, and a pattern/state variance ratio of about 2e-5. This is NASA PCoE's situation in v1, and the amplitudes are not inflated (D21).
- **Warnings (declared severity):** the generated $q_1$ IQR is 3.2–3.7 against 2.6 mAh, and the estimator check is 0.156 / 0.094 against 0.05.

## W5 RQ1 — fidelity of FORM, and TSTR without X3 on MATR and HUST

13 Oct · CUDA, 56 min · held-out units 55 (all reach EOL)
commit cc91851

| FORM INPUT SET | TRANSFER RATIO [95 % BCA] | RMSE TSTR / TRTR [CYCLES] |
|---|---|---|
| full | 1.539 [1.284, 1.742]; seed pairs 1.42–1.78 | 145 / 94 |
| capacity only | 1.083 [0.991, 1.146] | 147 / 135 |
| capacity relative | 1.121 [1.076, 1.167] | 138 / 123 |
| capacity state | 1.104 [1.057, 1.150] | 141 / 128 |
| full centred | 1.162 [1.053, 1.234] | 133 / 115 |

- **Transfer reading:** the interval excludes 1; the ratio lies outside HUST's interval, which it overlaps.
- **Where the gap sits.** Both readings hold under every leave-one-seed-out:

- full vs capacity only: 0.35 [0.17, 0.48];
- full vs full centred: 0.28 [0.10, 0.40].

Capacity relative does not hold (−0.03 [−0.10, 0.01]). FORM's v2 profile looks like MATR's v1 profile: the measured model gains from the early-life channel levels (135 → 94 cycles) and the generated-trained model does not (147 → 145).

- **Discriminator: readable** (D02b): 0.097 [0.059, 0.154], 0.103 [0.082, 0.124], 0.168 [0.122, 0.223]. X3 off: 0.087.
- **Covariance agreement** 0.35 against a measured 0.30 [0.20, 0.43]: closed. X3 off: 0.32.
- **Property table.**
  - Reproduced: drift 7.78 / 8.27 / 7.87 mAh per 100 cycles; T 701 / 665 / 683; offset IQRs; positive runs 0.33 / 0.29 / 0.27 per 100 positions.
  - Departing: capacity noise variance 1.43 → 0.99 mAh², and the $q_1$ IQR 2.63 → 3.43 mAh.
  - 279 of 300 generated units enter the EOL rows (see flags).

| TSTR WITHOUT X3 (DECLARED TSTR_WITHOUT_X3) | MATR | HUST |
|---|---|---|
| v1 (printed) / v1 full_D27 | 3.42 / 3.51 | 1.46 / 1.46 |
| v2 without X3, seed 0 | 1.60 [1.06, 2.50] | 1.12 [0.92, 1.29] |
| v2, seed 0 / three seeds | 1.58 / 1.49 | 1.15 / 1.17 |
| share of X2 (offsets) / X3 (correlated noise) | 2.19× [1.69, 2.79] / 1.02× [0.92, 1.10] | 1.30× [1.03, 1.60] / 0.98× [0.88, 1.06] |

The retrained seed-0 TRTR and TSTR models reproduce V5's per-unit errors exactly (relative difference 0).

## W6 RQ2 — usability of FORM, with the pattern term read for the first time

13 Oct · CUDA, 40 min · 10 members per weighting

commit 830ef2b

- **Gates pass:** NRMSE 0.179 / 0.183 (recency / uniform), 10/10 members; null leakage ≤ 0.0008.
- **X4:** every weighted channel is used by every member, materially. Conditional importance: capacity 0.146, charge time 0.094, mean discharge voltage 0.096 (lower bounds ≥ 0.074).
- **Offset share** 0.159 / 0.167; Spearman(y, RUL) −0.69.
- **Pattern term.**
  - Variance ratio 1.7e-5, against a band of 0.05–0.50.
  - Ablation 0.0000 [−0.0001, 0.0002], and 0.0003 [−0.0006, 0.0012] uniform: **not shown to be learned**. At a variance ratio of 1.7e-5 a learned term would change the error by about 1e-5, inside the interval.
  - Graded/sparse |r| 0.045 / 0.008, against a bound of 0.3. This check was declared but never computed in v2 and is added for profiles with patterns.

---

- By the declared reading (`v2_1.readings.sparse_set_trained`, D21 logic), FORM's trained-model sparse-set scores are void. The amplitudes are not inflated.

## W7 RQ3 — degraded maps on FORM, and paired retrieval

13 Oct · CPU, 110 s · 900 windows / 45 units; retrieval 640 / 32

commit 7b538bb

- **The registering magnitudes equal those of HUST** on every operator and score: noise 0.2 / 0.1 / 0.5; mass to start 1.0 / – / 0.05; mass to end 1.0 / – / 0.1; smoothing – / – / 8; permutation 0.2 / 0.1 / 0.2.
- **Chance (recency):** rank −0.001, allocation 0.476, temporal 5.21 positions.
- **Paired retrieval**, ported from v1 S7 into V7, run for the first time.
  - Undegraded 1.00 against a chance of 0.07.
  - Noise at 0.05 gives 0.37; permuting 5 % gives 0.29.
  - A shift of 40 % gives 0.74–0.75, and a full shift 0.10–0.11.
  - Smoothing at σ = 8 gives 0.993.

  Plain retrieval stays near chance (0.095), as D01 found.

## W8 Reference values on FORM, with paired maps

13 Oct · 12 CPU workers, 1.31 h · 4 680 TimeSHAP explanations

commit 8c049a6

- **Runner:** benchmark 1 920 windows/h at 12 workers, chunk 4096. Equivalence with the sequential path: max |diff| 0. Projection 2.44 h, within the 36 h ceiling.
- **Run:** 4 680 TimeSHAP jobs in 1.31 h (02:46–04:04). The 120 windows of 45 test units are each explained with their pattern-free counterpart. Scoring: 21 s on CUDA. 13 of 13 checks pass.
- **Reference model exact:** IG 4.6e-8, occlusion 1.4e-7, TimeSHAP ≤ 2.4e-5, against max |exact| 0.056. Ceiling 0.849 rank / 0.125 allocation / 0.63 positions.
- **Trained model:** rank 0.744–0.771, allocation 0.215–0.265, temporal 3.71–3.89 positions; zero-weight mass 0.039–0.048 (null channels).
- **Floor: no ordering in any FORM score.** Rank ranges are 0.32–0.34 against a method gap of 0.027. The allocation gap of 0.050 is below the IG and TimeSHAP ranges (0.061, 0.068). The ten FORM models disagree far more than those of MATR or HUST.
- **Paired retrieval**, on the 11 test units whose windows contain the sparse set: **1.00 on the reference model** for every method. On the trained model it is **void**: the pattern term is not shown to be learned (W6). The computed 0.82–0.94 only show that attribution changes where the input changes.
- **Stability:** the TimeSHAP seed spread is ≤ 0.0024 in rank. The D23 40-window subset holds in every cell.
- **Readouts:**
  - average event: TimeSHAP rank 0.771 → 0.740 (first to last, within the floor);
  - uniform weighting: trained rank falls to 0.53–0.57 against a ceiling of 0.826, the largest weighting effect of the three profiles, and temporal 3.63–4.25 lies above the permuted 1.77.

## W9 Methodology diagrams redrawn from scripts

13 Oct · 5 figures · paper/img/scripts/

commit 195ce3f

- Figures 1–5 are drawn by `fig_loop.py`, `fig_generator.py`, `fig_mean_component.py`, `fig_weights.py` and `fig_tstr.py` (new). A shared read-only loader feeds them. Every number and label is read from the declarations or from the fitted or generated FORM profile.
- The figures show the three profiles, per-unit offsets, correlated noise (r = −0.89), patterns in FORM only, recency as primary, and the five TSTR input sets.

- `fig_weights_r3.py` was removed: it would have overwritten the figure with the old L = 40 drawing.

## W10 Results, the highlighted paper, and provenance

13 Oct · 6 tables, 8 figures · AMENDMENTS round 8

commit the W10 commit on branch v2.1

- **Results.** `scripts/v2_1/w10_write_paper.py` writes `results/v2.1` (one JSON per table), the table bodies, Figures 6–8 and the v2.1 section of `docs/RESULTS_PROVENANCE.md`. MATR and HUST are read from the v2 artifacts.
- **Highlighting.** Every change to §1–§3 since `6221c50` is in `\rev` (methodology) or `\revw` (wording, lighter), with a margin note.
  - The mechanical check (`scripts/v2_1/w10_rev_coverage.py`) finds **0 uncovered spans, 42 pure deletions all next to a highlight, and 0 highlighted paragraphs without a note**.
  - It found six unhighlighted edits from earlier rounds and five paragraphs without notes. All are now marked.
- **§4–§5** are written new for three profiles, with a banner each. The Limitations post-data list is in `\rev`. The Abstract and Conclusion are drafted and marked. The Notes box sits on page 1 of the review build.
- **Consistency pass:** Three review lenses (numbers; consistency and claims; brief compliance), each followed by a refuter, reported 60 findings. The refuters rejected 14, confirmed 41 and corrected the fix of 5. **All 46 surviving findings (35 distinct after merging duplicates across lenses) were applied.** They include: the r4 timing now also names one cell's measured values; “not learned” became “not shown to be learned”; FORM's 63 formation groups; the discriminator statement of §3.4; the covariance claim limited to what was measured; and the formation-protocol share added to §3.4. The workflow journal is in the session; the findings and verdicts are summarised in the W10 review.
- **Builds** in `paper/review/`, as regular files: `DegradX_review.pdf`, `DegradX_clean.pdf` (no review markup, checked) and `latexdiff_vs_original.pdf` (against `6221c50`).
- **README.** Licence note: CC BY-NC 4.0 for FORM-derived parameters and units, with attribution to Cui *et al.* 2024.

## Decision and deviation records at a glance

| RECORD | STAGE | TIMING | WHAT IT SETTLES |
|---|---|---|---|
| D32 FORM extraction | W0 | pre-data, before download | documented cells only; aging cycles as positions; CC-CV charge time and its validation; rest events; pre-screen; archives; tolerance |
| D33 CC-phase charge current | W1 | POST-DATA | the protocol check reads the charge current on CC samples; no false second protocol |
| D34 FORM eligibility | W2 | readings declared in r4 | eligible; pre-screen 0.12; positive pattern type enabled, attributed to the diagnostic block |
| T7 ISU-ILCC run start | v2 V2, recorded at W0 | methodological, closed | mechanism window evaluated one position late; corrected for FORM, not recomputed for ISU-ILCC |

## What the programme establishes, and where it stops

The construction holds on all three profiles, and FORM tests it on a third chemistry, format and laboratory. The benchmark now carries an inserted pattern type, but at its fitted amplitude no trained model is shown to learn it.

### ESTABLISHED

- FORM is eligible by the rules declared before download. It is learnable and every channel is used.
- FORM's transfer gap (1.54) is located: in the non-capacity channels, mostly their early-life levels.
- Correlated noise has no share in transfer (MATR 1.02×, HUST 0.98×). The offsets carry the v2 gains.
- The covariance gap is closed on all three profiles.
- The registering pattern of the scores is identical on a third profile. Paired retrieval works as designed on degraded maps.
- On the reference model all three methods are exact on FORM too, and they localize its sparse set exactly (paired retrieval 1.00).

### VOID OR UNSUPPORTED

- Trained-model sparse-set scores on FORM (pattern term not shown to be learned).
- The transfer ratio of FORM with independent noise (declared for MATR and HUST only).
- The discriminator cells of MATR and HUST (D02b).
- MATR or FORM inside HUST's transfer interval.
- Any method ordering on FORM's trained models: the ten-model floor exceeds every gap.

### OPEN FOR THE AUTHORS

- The licence term (NC) for the release.
- The post-data rules D33 and the mechanism sensitivity; the r4 timing wording.
- Whether FORM's sparse set is presented as a construction check only.
- The AI-use disclosure, the Abstract and Conclusion drafts, the repository footnote.
- The candidate for the remaining gaps of MATR and FORM: channel trajectories whose shape differs between protocols. Mappings per protocol would test it, at the cost the offset bound governs.

---

Written from the committed v2.1 artifacts of the DegradX repository (branch `v2.1`):

- the stage reviews `artifacts/v2.1/*/REVIEW.md`;
- the decision records `docs/DECISIONS/D32–D34` and `docs/DEVIATIONS.md` (T7);
- `docs/AMENDMENTS.md` round 8 and `docs/RESULTS_PROVENANCE.md` (v2.1 section);
- `configs/declarations.yaml` (r4);
- `git log` on branch `v2.1`.

MATR and HUST numbers are those of tag `v2-results`. The v2 log is `docs/EXPERIMENT_LOG_v2.html`. This page has no figures; the v2.1 results figures are Figures 6–8 of the paper, in `results/v2.1/`.